Lab 7

Knowledge Representation in AI

- **Knowledge Representation (KR)** is a field in Artificial Intelligence (AI) that focuses on how knowledge can be **stored, structured, and used by computers** to make intelligent decisions.
- It allows machines to **reason, infer, and understand relationships** between different concepts.

Types of Knowledge in AI

1. **Declarative Knowledge:** Facts and information about the world.
    - Example: "Water is a liquid."
2. **Procedural Knowledge:** Step-by-step instructions to solve a problem.
    - Example: "To make tea, boil water, add tea leaves, and steep for 5 minutes."
3. **Semantic Knowledge:** Meaning and relationships between concepts.
    - Example: "A cat is a type of animal."
4. **Episodic Knowledge:** Past experiences and events.
    - Example: "I visited Paris in 2020."
5. **Metaknowledge:** Knowledge about knowledge.
    - Example: "I know that I don't know everything."

Methods of Knowledge Representation

1. Logical Representation

- Uses **logic-based rules** to describe knowledge.
- **Propositional Logic** and **First-Order Logic (FOL)** are common methods.

Example: Propositional Logic

- "If it is raining, then the ground is wet."
  raining -> wet_ground.

Example: First-Order Logic (FOL)

- "All humans are mortal."
  human(X) -> mortal(X).

2. Semantic Networks

- Uses **graphs** to represent relationships between concepts.
- **Nodes = Entities** (e.g., "Dog"), **Edges = Relationships** (e.g., "is a type of").

Example: Semantic Network

[Dog] ----> (is a) ----> [Animal]
[Dog] ----> (has) ----> [Four Legs]
[Dog] ----> (can) ----> [Bark]

- Used in **Natural Language Processing (NLP)** and **Ontology Modeling**.

3. Frames (Object-Oriented Representation)

- Represents **knowledge in structured "frames"** (like objects in Object-Oriented Programming).
- Each frame contains **attributes (slots) and values**.

Example: Frame-Based Representation of a Car

```
{
  "Car": {
    "Type": "Sedan",
    "Brand": "Toyota",
    "Color": "Red",
    "Wheels": 4
  }
}
```

- Used in **AI Chatbots and Recommendation Systems**.

4. Production Rules (If-Then Rules)

- Knowledge is represented in **IF-THEN** rules.
- Used in **Expert Systems**.

Example: Medical Diagnosis System

IF fever AND cough THEN flu.
IF sore_throat AND headache THEN cold.

5. Ontologies (Formal Knowledge Representation)

- A **structured way** to define relationships between concepts.
- Used in **Semantic Web, AI Assistants, and Search Engines**.

Example: Ontology in Healthcare

[Disease] ----> (is_a) ----> [Flu]
[Flu] ----> (has_symptom) ----> [Fever]
[Flu] ----> (treated_with) ----> [Paracetamol]

- Used in **Google's Knowledge Graph** and **IBM Watson**.

Knowledge Representation in Prolog

Prolog is a **logic-based AI language** used for **Knowledge Representation and Reasoning**.

1. Defining Facts and Rules

```
% Facts
human(socrates).
human(plato).
mortal(X) :- human(X).

% Query
?- mortal(socrates). % Output: true
```

2. Expert System Example (Animal Classification)

```
animal(tiger).
animal(eagle).
animal(dolphin).

mammal(X) :- animal(X), not(bird(X)).
bird(X) :- animal(X), has_feathers(X).

?- mammal(tiger). % Output: true
?- bird(eagle).   % Output: true
```

Integrating Knowledge Representation with APIs

- AI applications use **APIs** to fetch **real-world knowledge**.

Example: Weather API for Decision-Making

```
raining(london).
need_umbrella(X) :- raining(X).

?- need_umbrella(london). % Output: true
```

What is an API?

- API (**Application Programming Interface**) is a set of rules that allow different software applications to communicate.
- **Types of APIs:**
  - **Web APIs:** RESTful APIs (e.g., JSON, XML)
  - **Library APIs:** Functions provided by software libraries
  - **Operating System APIs:** Interfaces for system operations

How to Use an API?

- **Make a request to an API** using HTTP methods:
  - **GET**: Retrieve data
  - **POST**: Send data
  - **PUT**: Update data
  - **DELETE**: Remove data

Example: Using Python Requests Library

```
import requests

url = "https://api.example.com/data"
response = requests.get(url)

if response.status_code == 200:
    data = response.json()
    print(data)
else:
    print("Error fetching data")
```

What is an API Format?

- APIs return data in formats such as:
  - **JSON (JavaScript Object Notation) – Most common**

    ```
    {"name": "Alice", "age": 25, "city": "New York"}
    ```

  - **XML (Extensible Markup Language)**

    ```
    <person>
      <name>Alice</name>
      <age>25</age>
      <city>New York</city>
    </person>
    ```

Using an API in a Flask Web App

Steps to Integrate an API into Flask:

1. Install Flask
2. pip install flask requests
3. Create a Flask App
4. Fetch API Data
5. Display Data in HTML Front-End

Lab 7 - Tasks

Task 1; Create any of the following, can use Free API & develop its "Flask" based back-end only:

1. Weather - App
2. Cooking / Recipe - App
3. Spotify - App
4. News - App
5. Vehicle Info - App
6. Useless Info - App
7. Random Joke - App
8. Stock Price - App
9. NASA - App
10. Your Own Choice - App